

베이지안 기계학습

변분 추론의 기본 개념

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Introduction

Motivation I

- Bayesians infer d -dimensional parameter θ using a posterior density obtained from Bayes theorem

$$p(\theta | y) = \frac{p(\theta)p(y | \theta)}{p(y)} \propto p(\theta)p(y | \theta),$$

where $p(\theta)$ is a prior density and $p(y | \theta)$ is a likelihood.

- When $p(\theta | y)$ is intractable, we draw posterior samples using Markov chain Monte Carlo (MCMC) methods.

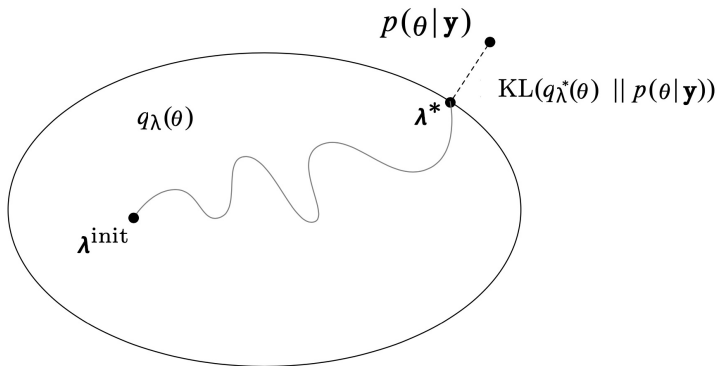
Motivation II

- ▶ When a dataset is large or likelihood evaluations are expensive for other reasons, there is an interest in approximate inference methods which have reduced computational demands compared to MCMC.

Variational Methods I

- ▶ Variational methods are deterministic approximate inference algorithms.
- ▶ The basic idea is to pick an approximation $q_{\lambda}(\theta)$ to the distribution from some tractable family, and then to try to make this approximation as close as possible to the true posterior.

Variational Methods II



Variational Methods III

- The closeness is measured by (reverse) Kullback-Leibler divergence:

$$KL(q_{\lambda}(\theta) \parallel p(\theta|\mathbf{y})) = \int \log \frac{q_{\lambda}(\theta)}{p(\theta|\mathbf{y})} q_{\lambda}(\theta) d\theta.$$

Remark on the KL divergence

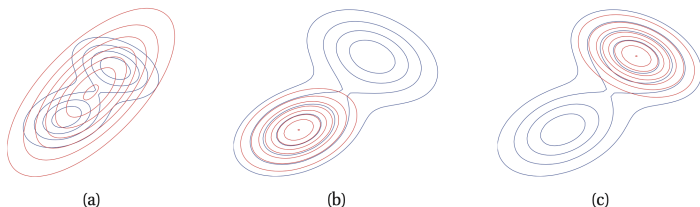


Figure: Machine Learning by Murphy (2010): The blue curves are the contours of the true posterior and the red curves are the contours of the unimodal approximation. (a) forwards KL. (b-c) reverse KL

- The (reverse) KL divergence is not only more tractable to compute, but also more sensible statistically.

The Evidence Lower Bound I

- Recall that the Kullback-Leibler divergence is

$$\begin{aligned}KL(q_\lambda(\theta) \parallel p(\theta|\mathbf{y})) &= \int \log \frac{q_\lambda(\theta)}{p(\theta|\mathbf{y})} q_\lambda(\theta) d\theta \\ &= - \int \log \frac{h(\theta)}{q_\lambda(\theta)} q_\lambda(\theta) d\theta + \log p(\mathbf{y})\end{aligned}$$

where $h(\theta) = p(\theta)p(\mathbf{y} \mid \theta)$.

The Evidence Lower Bound II

- In practice, we maximize the **evidence lower bound (ELBO)**:

$$\begin{aligned}\mathcal{L}(\lambda) &= \int \log \frac{h(\theta)}{q_\lambda(\theta)} q_\lambda(\theta) d\theta \\ &= \mathbb{E}_q[\log h(\theta)] - \mathbb{E}_q[\log q_\lambda(\theta)].\end{aligned}$$

Remarks on the ELBO

- ▶ It can be used to assess convergence of the algorithm.
- ▶ It can be used to assess the correctness of one's code: If the bound does not increase monotonically, there must be a bug.
- ▶ It can be used as an approximation to the marginal likelihood, which can be used for Bayesian model selection.

Coordinate ascent variational inference

Mean-field variational approximation

- The mean-field variational approximation (Opper and Saad, 2001) assume the posterior is a fully factorized approximation of the following form

$$q_{\lambda}(\theta) = \prod_j q_{\lambda_j}(\theta_j).$$

Coordinate ascent algorithm I

- ▶ The coordinate ascent algorithm (Ghahramani and Beal, 2001) maximizes $\mathcal{L}(\lambda)$ by updating each term in turn with all other terms fixed.
- ▶ The update for $q_{\lambda_j}(\theta_j)$ takes the form

$$q_{\lambda_j}(\theta_j) \propto \exp [\mathbb{E}_{-j} (\log p(\theta) p(\mathbf{y} \mid \theta))],$$

where $\mathbb{E}_{-j}(\cdot)$ denotes expectation with respect to $\prod_{l \neq j} q_{\lambda_l}(\theta_l)$.

Coordinate ascent algorithm II

Algorithm 1 Coordinate ascent variational inference (CAVI)

- 1: Initialize: Variational factors $q_{\lambda_j}(\theta_j)$
 - 2: **while** the ELBO has not converged **do**
 - 3: **for** $j \in \{1, \dots, d\}$ **do**
 - 4: Set $q_{\lambda_j}(\theta_j) \propto \exp [\mathbb{E}_{-j} (\log p(\theta) p(\mathbf{y} \mid \theta))]$
 - 5: **end for**
 - 6: Compute ELBO $\mathcal{L}(\lambda)$
 - 7: **end while**
 - 8: Return: $q_{\lambda}(\theta)$
-

Example: Gaussian regression model I

- Model:

$$y \sim \mathcal{N}\left(X\beta, \frac{1}{\tau}I_n\right),$$

where $y = (y_1, \dots, y_N)^\top$ and $X \in \mathbb{R}^{N \times p}$ is the design matrix.

- Priors:

$$\beta \sim \mathcal{N}\left(0, \frac{1}{\alpha}I_p\right), \quad \tau \sim \mathcal{Ga}(a_\tau, b_\tau).$$

- Variational distribution:

$$q_\lambda(\theta) = q_{\lambda_1}(\beta)q_{\lambda_2}(\tau),$$

where $\theta = (\beta, \tau)^\top$.

Example: Gaussian regression model II

- Joint posterior:

$$\log h(\theta) = \frac{N}{2} \log \tau - \frac{\tau}{2} \|y - X\beta\|^2 - \frac{\alpha}{2} \|\beta\|^2 + (a_\tau - 1) \log \tau - b_\tau \tau + C$$

Example: Gaussian regression model III

- Update β :

$$\begin{aligned}\mathbb{E}_{q(\tau)} \log h(\theta) &= -\frac{\mathbb{E}_{q(\tau)}(\tau)}{2} \|y - X\beta\|^2 - \frac{\alpha}{2} \|\beta\|^2 + C \\ &= (\beta - \beta_n)^\top V(\beta - \beta_n) + C,\end{aligned}$$

where $\beta_n = (\mathbb{E}_{q(\tau)}(\tau)) V^{-1} X^\top y$ and $V = (\mathbb{E}_{q(\tau)}(\tau)) X^\top X + \alpha I_p$.

- The variational distribution for β is given by

$$q_{\lambda_1}(\beta) = \mathcal{N}(\beta_n, V^{-1}),$$

where $\lambda_1 = (\beta_n, V)$.

Example: Gaussian regression model IV

- Update τ :

$$\begin{aligned}\mathbb{E}_{q(\beta)} \log h(\theta) &= \frac{N}{2} \log \tau - \frac{\tau}{2} \mathbb{E}_{q(\beta)} \|y - X\beta\|^2 + (a_\tau - 1) \log \tau - b_\tau \tau + C \\ &= \left(\frac{N}{2} + a_\tau - 1 \right) \log \tau - \left(\frac{1}{2} \mathbb{E}_{q(\beta)} \|y - X\beta\|^2 + b_\tau \right) \tau + C,\end{aligned}$$

where $\beta_n = (\mathbb{E}_{q(\tau)}(\tau)) V^{-1} X^\top y$ and $V = (\mathbb{E}_{q(\tau)}(\tau)) X^\top X + \alpha I_p$.

- The variational distribution for τ is given by

$$q_{\lambda_2}(\tau) = \mathcal{Ga} \left(a_\tau + \frac{N}{2}, b_\tau + \frac{1}{2} \mathbb{E}_{q(\beta)} \|y - X\beta\|^2 \right),$$

where $\lambda_2 = (a_\tau + N/2, \mathbb{E}_{q(\beta)} \|y - X\beta\|^2/2)$.

Stochastic gradient ascent variational inference

Gaussian variational approximation

- ▶ The Gaussian variational approximation uses a correlated Gaussian distribution:

$$q_{\lambda}(\theta) = \mathcal{N}(\mu, \Sigma), \quad \Sigma = LL^{\top},$$

where L is the Cholesky factor and the variational parameters $\lambda = (\mu, L)^{\top}$.

- ▶ Remarks:
 - ▶ This transforms the support of the parameters θ to the real coordinate space before choosing an approximation.
 - ▶ Full covariance is richest, but computationally expensive.

Stochastic gradient ascent algorithm I

- The ELBO $\mathcal{L}(\lambda)$ can be maximized by a stochastic gradient ascent algorithm:

$$\lambda^{(t+1)} = \lambda^{(t)} + \gamma_t \widehat{\nabla}_{\lambda} \mathcal{L} \left(\lambda^{(t)} \right),$$

where $\widehat{\nabla}_{\lambda} \mathcal{L}(\lambda)$ is an unbiased gradient and γ_t , $t \geq 1$, is a learning rate sequence satisfying the Robbins-Monro condition (Robbins and Monro, 1951)

$$\sum_t \gamma_t = \infty, \quad \sum_t \gamma_t^2 < \infty.$$

- Learning Rate: AdaGrad (Duchi et al., 2011); AdaDelta (Zeiler, 2012); ADAM (Kingma and Ba, 2017)

Stochastic gradient ascent algorithm II

- Reparametrization trick (Kingma and Welling, 2013):

$$\begin{aligned}\mathcal{L}(\lambda) &= \mathbb{E}_q(\log h(\theta) - \log q(\theta \mid \mu, L)) \\ &= \mathbb{E}_f(\log h(\mu + Ls) - \log q(\mu + Ls \mid \mu, L)) \\ &= \mathbb{E}_f(\log h(\mu + Ls)) + \log |L| + K,\end{aligned}$$

where $s = L^{-1}(\theta - \mu)$ is distributed according to the density f (standard normal) and K denotes a term not depending on μ, L .

- The lower bound can be rewritten as an expectation with respect to a fixed density f that does not depend on the variational parameters.

Stochastic gradient ascent algorithm III

- Derivative of $\mathcal{L}(\lambda)$ with respect to μ :

$$\nabla_{\mu} \mathbb{E}_f(\log h(\mu + Ls)) = \mathbb{E}_f(\nabla_{\theta} \log h(\mu + Ls)).$$

Stochastic gradient ascent algorithm IV

- Derivative of $\mathcal{L}(\lambda)$ with respect to L :

$$\begin{aligned}\nabla_{\text{vec}(L)} \mathbb{E}_f(\log h(\mu + Ls)) &= \nabla_{\text{vec}(L)} \mathbb{E}_f(\log h(\mu + (s^\top \otimes I) \text{vec}(L))) \\ &= \mathbb{E}_f((s \otimes I) \nabla_\theta \log h(\mu + Ls)) \\ &= \mathbb{E}_f(\text{vec}(\nabla_\theta \log h(\mu + Ls) s^\top)). \\ \nabla_L \mathbb{E}_f(\log h(\mu + Ls)) &= \mathbb{E}_f(\nabla_\theta \log h(\mu + Ls) s^\top).\end{aligned}$$

- Remarks:

- $\nabla_A g(A)$ means $\text{vec}^{-1}(\nabla_{(A)} g(A))$.
- $\text{vec}(ABC) = (C^\top \otimes A) \text{vec}(B)$.

Stochastic gradient ascent algorithm V

Algorithm 2 Stochastic gradient ascent variational inference (SGDVI)

- 1: Initialize: $\mu^{(0)} = 0$ and $L^{(0)} = I_d$
 - 2: **while** the ELBO has not converged **do**
 - 3: Draw $s \sim \mathcal{N}(0, I)$
 - 4: Compute $\theta^{(t)} = \mu^{(t)} + L^{(t)} s$
 - 5: Update step-size γ_t
 - 6: Update $\mu^{(t+1)} = \mu^{(t)} + \gamma_t \nabla_{\theta} \log h(\theta^{(t)})$
 - 7: Update $L^{(t+1)} = L^{(t)} + \gamma_t \left\{ \nabla_{\theta} \log h(\theta^{(t)}) s^{\top} + \text{diag}(1/L_{11}^{(t)}, \dots, 1/L_{dd}^{(t)}) \right\}$
 - 8: Compute ELBO $\mathcal{L}(\lambda)$
 - 9: **end while**
 - 10: Return: μ^* and L^*
-

Example: Gaussian regression model I

- Model:

$$y \sim \mathcal{N}\left(X\beta, \frac{1}{\tau}I_n\right),$$

where $y = (y_1, \dots, y_N)^\top$ and $X \in \mathbb{R}^{N \times p}$ is the design matrix.

- Priors:

$$\beta \sim \mathcal{N}\left(0, \frac{1}{\alpha}I_p\right), \quad \tau \sim \mathcal{Ga}(a_\tau, b_\tau).$$

- Variational distribution:

$$q_\lambda(\theta) = \mathcal{N}(\theta \mid \mu, LL^\top),$$

where $\theta = (\beta, \log \tau)^\top$ and $\lambda = (\mu, L)$.

Example: Gaussian regression model II

- Joint posterior:

$$\log h(\theta) = \frac{N}{2} \log \tau - \frac{\tau}{2} \|y - X\beta\|^2 - \frac{\alpha}{2} \|\beta\|^2 + (a_\tau - 1) \log \tau - b_\tau \tau + C$$

Example: Gaussian regression model III

- Compute gradients:

$$\nabla_{\beta} \log h(\theta) = -e^{\phi} X^{\top} (y - X\beta) - \alpha\beta,$$

$$\nabla_{\phi} \log h(\theta) = \frac{N}{2} - \frac{e^{\phi}}{2} \|y - X\beta\|^2 + (a_{\lambda} - 1) - b_{\lambda} e^{\phi},$$

where $\phi = \log \tau$.

Black box variational inference

Stochastic gradient ascent algorithm VI

- Log-derivative trick ($\nabla_{\theta} \log h(\theta)$ is not available):

$$\begin{aligned}\nabla_{\lambda} \mathcal{L}(\lambda) &= \nabla_{\lambda} \int \{\log h(\theta) - \log q_{\lambda}(\theta)\} q_{\lambda}(\theta) d\theta \\&= \int \{\log h(\theta) - \log q_{\lambda}(\theta)\} \nabla_{\lambda} \log q_{\lambda}(\theta) q_{\lambda}(\theta) d\theta \\&\quad - \int \nabla_{\lambda} \log q_{\lambda}(\theta) q_{\lambda}(\theta) d\theta \\&= \mathbb{E}_q(\{\log h(\theta) - \log q_{\lambda}(\theta)\} \nabla_{\lambda} \log q_{\lambda}(\theta))\end{aligned}$$

where $\mathbb{E}(\nabla_{\lambda} \log q_{\lambda}(\theta)) = 0$ and $\nabla_{\lambda} q_{\lambda}(\theta) = \nabla_{\lambda} \log q_{\lambda}(\theta) q_{\lambda}(\theta)$.

Stochastic gradient ascent algorithm VII

- Naive estimate of $\nabla_{\lambda} \mathcal{L}(\lambda)$:

$$\nabla_{\lambda} \mathcal{L}(\lambda) \approx \frac{1}{S} \sum_{s=1}^S \{\log h(\theta^{(s)}) - \log q_{\lambda}(\theta^{(s)})\} \nabla_{\lambda} \log q_{\lambda}(\theta^{(s)}),$$

where $\theta^{(1)}, \dots, \theta^{(S)} \stackrel{iid}{\sim} q_{\lambda}(\theta)$.

Stochastic gradient ascent algorithm VIII

- Control variate:

$$\nabla_{\lambda} \mathcal{L}(\lambda) \approx \frac{1}{S} \sum_{s=1}^S \{\log h(\theta^{(s)}) - \log q_{\lambda}(\theta^{(s)}) - z\} \nabla_{\lambda} \log q_{\lambda}(\theta^{(s)}),$$

where z is an arbitrary constant and its best choice is given by

$$z = \frac{\text{Cov}((\log h(\theta) - \log q_{\lambda}(\theta)) \nabla_{\lambda}(\log q_{\lambda}(\theta)), \nabla_{\lambda}(\log q_{\lambda}(\theta)))}{\text{Var}(\nabla_{\lambda}(\log q_{\lambda}(\theta)))}$$

Stochastic gradient ascent algorithm IX

Algorithm 3 Black-box variational inference (BBVI)

- 1: Initialize: $\lambda^{(0)}$ randomly
 - 2: **while** the ELBO has not converged **do**
 - 3: Draw $\theta^{(1)}, \dots, \theta^{(S)} \sim q_{\lambda}(\theta)$
 - 4: Compute $\hat{z}$
 - 5: Compute $\hat{\nabla}_{\lambda} \mathcal{L}(\lambda)$
 - 6: Update step-size γ_t
 - 7: Update $\lambda^{(t+1)} = \lambda^{(t)} + \gamma_t \hat{\nabla}_{\lambda} \mathcal{L}(\lambda)$
 - 8: Compute ELBO $\mathcal{L}(\lambda)$
 - 9: **end while**
 - 10: Return: λ^*
-

Example: Gaussian regression model I

- Model:

$$y \sim \mathcal{N}\left(X\beta, \frac{1}{\tau}I_n\right),$$

where $y = (y_1, \dots, y_N)^\top$ and $X \in \mathbb{R}^{N \times p}$ is the design matrix.

- Priors:

$$\beta \sim \mathcal{N}\left(0, \frac{1}{\alpha}I_p\right), \quad \tau \sim \mathcal{Ga}(a_\tau, b_\tau).$$

- Variational distribution:

$$q_\lambda(\theta) = \mathcal{N}(\beta \mid \mu_{N,\beta}, \sigma_{N,\beta}^2) \mathcal{Ga}(\tau \mid a_{N,\tau}, b_{N,\tau}),$$

where $\theta = (\beta, \tau)^\top$ and $\lambda = (\mu_{N,\beta}, \sigma_{N,\beta}^2, a_{N,\tau}, b_{N,\tau})$.

Example: Gaussian regression model II

- Joint posterior:

$$\log h(\theta) = \frac{N}{2} \log \tau - \frac{\tau}{2} \|y - X\beta\|^2 - \frac{\alpha}{2} \|\beta\|^2 + (a_\tau - 1) \log \tau - b_\tau \tau + C_1$$

Example: Gaussian regression model III

- Variational distribution:

$$\begin{aligned}\log q_{\lambda}(\theta) = & -\frac{p}{2} \log \sigma_{N,\beta}^2 - \frac{1}{2\sigma_{N,\beta}^2} \|\beta - \mu_{N,\beta}\|^2 + C_2 \\ & + a_{N,\tau} \log b_{N,\tau} - \log \Gamma(a_{N,\tau}) + (a_{N,\beta} - 1) \log \tau - b_{N,\beta} \tau\end{aligned}$$

Example: Gaussian regression model IV

- Compute gradients:

$$\nabla_{\mu_{N,\beta}} \log q_{\lambda}(\theta) = \frac{1}{\sigma_{N,\beta}^2} (\beta - \mu_{N,\beta}),$$

$$\nabla_{\sigma_{N,\beta}^2} \log q_{\lambda}(\theta) = -\frac{p}{2\sigma_{N,\beta}^2} + \frac{1}{2\sigma_{N,\beta}^4} \|\beta - \mu_{N,\beta}\|^2,$$

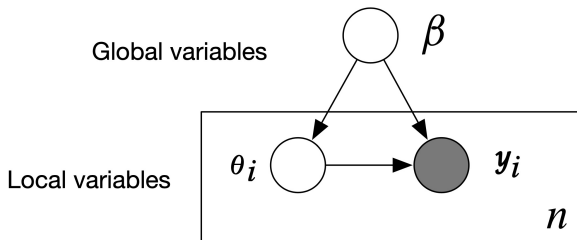
$$\nabla_{a_{N,\tau}} \log q_{\lambda}(\theta) = \log b_{N,\tau} - \frac{\Gamma'(a_{N,\tau})}{\Gamma(a_{N,\tau})} - \log \tau,$$

$$\nabla_{a_{N,\tau}} \log q_{\lambda}(\theta) = \frac{a_{N,\tau}}{b_{N,\tau}} - \tau.$$

Stochastic variational inference

Models with local and global parameters I

► Graphical model:



$$p(\beta, \theta, \mathbf{y}) = p(\beta) \prod_{i=1}^n p(\theta_i, y_i | \beta)$$

Models with local and global parameters II

- Consider a longitudinal generalized linear mixed model (GLMM) where the response $y_i = (y_{i1}, \dots, y_{in_i})^\top$ is a member of the exponential family distribution and the link function is given by

$$g(\mu_{ij}) = X_{ij}\beta + Z_{ij}b_i, \quad i = 1, \dots, N, \quad j = 1, \dots, n_i, \\ b_i \sim \mathcal{N}(0, \Sigma_R), \quad i = 1, \dots, N,$$

where X_{ij} and Z_{ij} are vectors of covariates, β is a vector of fixed effects parameters and b_i is a random effect for the i th subject.

Stochastic variational inference I

► Key idea:

1. Randomly select a mini-batch B of $|B| \geq 1$ units from the whole data set.
2. Optimize local variational parameters of units in mini-batch B (as a function of the global variational parameters at their current setting).
3. Update global variational parameters using stochastic natural gradient ascent. Noisy gradients are computed based on optimized local variational parameters of units in mini-batch B .

Stochastic variational inference II

- ▶ The ELBO $\mathcal{L}(\lambda)$ for global variational parameters can be maximized by a stochastic natural gradient ascent algorithm:

$$\lambda^{(t+1)} = \lambda^{(t)} + \gamma_t I_F^{-1}(\lambda) \widehat{\nabla}_{\lambda} \mathcal{L} \left(\lambda^{(t)} \right),$$

where $I_F^{-1}(\lambda) = \text{Cov}(\nabla_{\lambda} \log q_{\lambda}(\theta))$ is the Fisher information matrix about λ with respect to the distribution $q_{\lambda}(\theta)$.

- ▶ Remarks:
 - ▶ Amari (1982, 1998) showed that the natural gradient accounts for the geometric structure of probability parameters.

Remark and next week

Remark

- ▶ When a dataset is large or likelihood evaluations are expensive, there is an interest in approximate inference methods which have reduced computational demands compared to MCMC.
- ▶ Variational methods are widely used to approximate the true posterior.
- ▶ The basic idea is to pick an approximation $q_{\lambda}(\theta)$ to the distribution from some tractable family, and then to try to make this approximation as close as possible to the true posterior.

Next week

- ▶ Bayesian inference in NumPyro
 - ▶ Markov chain Monte Carlo
 - ▶ Variational inference

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